

REVIEW

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AI literacy in K-12: a systematic literature review

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Abstract

The successful irruption of AI-based technology in our daily lives has led to a growing educational, social, and political interest in training citizens in AI. Education systems now need to train students at the K-12 level to live in a society where they must interact with AI. Thus, AI literacy is a pedagogical and cognitive challenge at the K-12 level. This study aimed to understand how AI is being integrated into K-12 education worldwide. We conducted a search process following the systematic literature review method using Scopus. 179 documents were reviewed, and two broad groups of AI literacy approaches were identified, namely learning experience and theoretical perspective. The first group covered experiences in learning technical, conceptual and applied skills in a particular domain of interest. The second group revealed that significant efforts are being made to design models that frame AI literacy proposals. There were hardly any experiences that assessed whether students understood AI concepts after the learning experience. Little attention has been paid to the undesirable consequences of an indiscriminate and insufficiently thought-out application of AI. A competency framework is required to guide the didactic proposals designed by educational institutions and define a curriculum reflecting the sequence and academic continuity, which should be modular, personalized and adjusted to the conditions of the schools. Finally, AI literacy can be leveraged to enhance the learning of disciplinary core subjects by integrating AI into the teaching process of those subjects, provided the curriculum is co-designed with teachers.

Keywords Secondary education, Teaching/learning strategies, Twenty-first century skills, Cultural and social implications

Introduction

In recent years, the convergence of huge computing power, massive amounts of data and improved machine learning algorithms have led to remarkable advances in Artificial Intelligence (AI) based technologies, which are

set to be the most socially and economically disruptive technologies ever developed (Russell, 2021). The irruption of AI-based technology in our daily lives (e.g., robot vacuum cleaners, real-time location and search systems, virtual assistants, etc.) has generated a growing social and political interest in educating citizens about AI. The scientific community has also begun to engage in this education after detecting a significant gap in the understanding of AI, based on comments and fears expressed by citizens about this technology (West & Allen, 2018). Therefore, integrating AI into curricula is necessary to train citizens who must increasingly live and act in a world with a significant presence of AI.

It is worth noting that AI education addresses not only the learning of the scientific and technological

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foundations of AI, but also the knowledge and critical reflection on how a trustworthy AI should be developed and the consequences of not doing so. Hence, it is crucial to incorporate AI teaching from the earliest stages of education (Heintz, 2021). However, although some countries are making significant efforts to promote AI teaching in K-12 (Touretzky et al., 2019a), this is being implemented through highly varied AI training experiences, such as data-driven design (Vartiainen et al., 2021), interactive data visualizations (Chittora & Baynes, 2020; von Wangenheim et al., 2021), virtual reality and robotics (Narahara & Kobayashi, 2018), games (Giannakos et al., 2020), or even based on combined workshop series (Lee et al., 2021). To date, there are very few methodological proposals on how to introduce the AI curriculum in K-12 education (Lee et al., 2020).

Since the development of a field requires prior research, we propose in this paper to identify and examine the way in which AI literacy is developing in K-12 around the world, to draw conclusions and guide teaching proposals for AI literacy in K-12. By highlighting and discussing the pros and cons of the different approaches and experiences in the literature, we aim to inspire new initiatives and guide the actors involved, from decisions-makers, for example in education policy, to teachers involved in their conception, design and implementation. We also hope to raise awareness of the importance of learning about AI from an early age, emphasizing the key aspects of this training and, hopefully, fueling the debate that needs to be fostered in our research community.

Integration of AI into the K12 curriculum

As a scientific-technological field, AI is just a few decades old. The name was coined in 1956, and since then different disciplines (such as computer science, mathematics, philosophy, neuroscience, or psychology) have contributed to its development from an interdisciplinary focus. AI is oriented to comprehend, model, and replicate human intelligence and cognitive processes into artificial systems. Currently, it covers a wide range of subfields such as machine learning, perception, natural language processing, knowledge representation and reasoning, computer vision, among many others (Russell & Norvig, 2021).

Starting in the 1970s, AI began to emerge in educational contexts through tools specifically designed to support learning, teaching, and the management of educational institutions. Since many jobs are now AI-related and will continue to increase in the coming years, some researchers believe that AI education should be considered as important as literacy in reading and writing (Kandlhofer et al., 2016). The highly interdisciplinary character is also another factor to consider. AI literacy

can be defined as a set of skills that enable a solid understanding of AI through three priority axes: learning about AI, learning about how AI works, and learning for life with AI (Long & Magerko, 2020; Miao et al., 2021). The first axis focuses on understanding AI concepts and techniques to enable the recognition of which artifacts/platforms use AI and which do not. The second axis addresses the understanding of how AI works, to effectively interact with it. The third axis seeks to understand how AI can affect our lives, allowing us to critically evaluate its technology. Thus, AI literacy goes beyond the use of AI applications in education, such as Intelligent Tutoring Systems (ITS) (du Boulay, 2016).

The teaching of knowledge in AI has traditionally been carried out at the university level, focused on students who study disciplines closely related to computing and ICT in general. In recent years, AI learning has also started to be relevant both in university programs with diverse study backgrounds (Kong et al., 2021), as well as at the K-12 level (Kandlhofer & Steinbauer, 2021; Tedre et al., 2021). However, teaching AI at the K-12 level is not yet prevalent in formal settings and is considered challenging. Experts believe it is important to have some thought on what AI education should look like at the K-12 level so that future generations can become informed citizens who understand the technologies they interact with in their daily lives (Touretzky et al., 2019a). Training children and teenagers will allow them to understand the basics of the science and technology that underpins AI, its possibilities, its limits and its potential social and economic impact. It also stimulates and better prepares them to pursue further studies related to AI or even to become creators and developers of AI themselves (Heintz, 2021).

Nowadays, research on AI teaching is still scarce (Chai et al., 2020a, 2020b; Lee et al., 2020). The acquisition of knowledge in AI represents a great pedagogical challenge for both experts and teachers, and a cognitive challenge for students (Micheuz, 2020). Some countries are making significant efforts to promote AI education in K-12 (Touretzky et al., 2019b), by developing relatively comprehensive curriculum guidelines (Yue et al., 2021). Through interviews with practitioners and policy makers from three different continents (America, Asia and Europe), some studies report on continuing works to introduce AI in K-12 education (He et al., 2020). Some other work focuses on examining and comparing AI curricula in several countries (Yue et al., 2021). In addition, there are a growing number of AI training experiences that explore pathways to optimize AI learning for K-12 students. However, most of them are somehow thematically limited as they do not adequately address key areas of AI, such as planning,

knowledge representation and automated reasoning (Nisheva-Pavlova, 2021). Additionally, due to the rapid growth of AI, there is a need to understand how educators can best leverage AI techniques for the academic success of their students. Zhai et al. (2021) recommend that educators work together with AI experts to bridge the gap between technique and pedagogy.

Using a systematic review method, our research aims to present an overview of current approaches to understand how AI is taught worldwide. Several studies have conducted systematic reviews concerning applications of AI in education. Zhai et al. (2021) analyzed how AI was applied to the education domain from 2010 to 2020. Their review covers research on AI-based learning environments, from their construction to their application and integration in the educational environment. Guan et al. (2020) reviewed the main themes and trends in AI research in education over the past two decades. The authors found that research on the use of AI techniques to support teaching or learning has stood the test of time and that learner profiling models and learning analytics have proliferated in the last two decades. Ng et al. (2022) examined learner types, teaching tools and pedagogical approaches in AI teaching and learning, mainly in university computer science education. Chen et al. (2020) covered education enhanced by AI techniques aimed to back up teaching and learning. All these studies have focused on the main role that AI has played in educational applications over the last decades. However, in light of the recent need to consider how AI education should be approached at the K-12 level (Kandlhofer et al., 2016; Long & Magerko, 2020; Miao et al., 2021; Touretzky et al., 2019b), it would be of great value to structure and characterize the different approaches used so far to develop AI literacy in K-12, as well as to identify research gaps to be explored. Recently, Yue et al. (2022) analyzed the main components of the pedagogical design in 32 empirical studies in K-12 AI education and Su et al. (2022) examined 14 learning experiences carried out in the Asian-Pacific region. These components included target audience, setting, duration, contents, pedagogical approaches to teaching, and assessment methods. Sanusi et al. (2022) reviewed research on teaching machine learning in K-12 from four perspectives: curriculum development, technology development, pedagogical development, and teacher training development. The findings of the study revealed that more studies are needed on how to integrate machine learning into subjects other than computer science. Crompton et al. (2022) carried out a systematic review on the use of AI as a supporting tool in K-12 teaching, which entails an interesting but

narrower scope. Our study extends previous reviews on K-12 AI research by emphasizing how the current approaches are integrating AI literacy in K-12 education worldwide.

Research question

To begin the systematic review, a single research question (RQ) was formulated.

RQ: How are current approaches integrating AI literacy into K-12 education worldwide?

In essence, the RQ aims to investigate the characterization of the different approaches being employed to incorporate AI education in K-12. The following subsections in the methodology describe the search and the data collection process in such a way that an answer to the RQ can be provided in a replicable and objective fashion.

Methods

The research method chosen to conduct this research was the systematic literature review (SLR), following the guidelines posed by Kitchenham (2004). Accordingly, the following subsections summarize and document the key steps implemented in this research method.

Search process

We used *Scopus* to implement the search process. *Scopus* provides an integrated search facility to find relevant papers in its database based on curated metadata. It includes primary bibliographic sources published by *Elsevier*, *Springer*, *ACM*, and *IEEE*, among others. It provides a comprehensive coverage of journals and top-ranked conferences within fields of interest. We did not limit our search to specific journals or regular conference proceedings, as there is not yet a clearly established body of literature on the subject. All searches were performed based on title, keywords and abstract, and conducted between 21 October 2021 and 9 March 2023.

To decide the search string, we ran an initial search and found only a few papers focused on 'literacy' whereas the vast majority referred to the broader term 'education'. Therefore, we decided to use both search terms (key issue 1 in Table 1). As some recent works combine the terms 'Artificial Intelligence' and 'education'/'literacy' into single terms such as 'AI literacy' or 'AI education', these were added to the search string (key issue 2 in Table 1). The educational stage was also included in the search string (key issue 3 in Table 1). As the search term 'education' also returns AI-based learning environments which are outside the scope of our review, we explicitly considered negated terms to leave out both computer-based learning and intelligent tutoring systems (key issue 4 in Table 1). A final decision was whether to use the term 'Artificial

Table 1 Summary of search terms and issues associated with their choice

Key issues	Search
1 Education vs. literacy	literacy OR education
2 Artificial Intelligence & Education/literacy	((literacy OR education) AND ((artificial AND intelligence))) OR ("AI literacy" OR "AI education")
3 Educational stage	("primary school" OR "secondary school" OR k-12 OR "middle school" OR "high school")
4 Excluding Ai-based learning environments	Excluding Ai-based learning environments & NOT ("computer-based learning") AND NOT ("intelligent tutoring system")

Table 2 Data extraction annotations definitions

Annotations on extracted data	Definition
Study metadata information	Type of publication (journal or proceedings), publication details, primary study ID, author(s), title, abstract
Type of educational approach	Learning experience, theoretical perspective
Interdisciplinary nature	Yes, No
AI support tools	Yes (the study includes learning and acquiring skills in AI support tools), No
Student role	Yes (the study describes the role of the student in the AI education process), No
Teacher training	Yes (the study describes the training of teachers and their role in the teaching and learning process), No
Systemic experience	Yes (The study means a systemic experience/proposal rooted in country-wide education policies), No

Intelligence' as a single umbrella term or to add narrower terms related to AI subfields (e.g., machine learning). After a preliminary inspection of a few relevant papers, we observed that such additional specific terms usually co-occur with the string 'Artificial Intelligence' in education, and they were therefore regarded as unnecessary. Thus, to capture the essence of our RQ and to build up the complete search string, we considered the search terms as shown in Table 1. Eventually, this resulted in the following complete search string in Scopus:

TITLE-ABS-KEY (((literacy OR education) AND ((artificial AND intelligence))) OR ("AI literacy" OR "AI education")) AND ("primary school" OR "secondary school" OR k-12 OR "middle school"

OR "high school") AND NOT ("computer-based learning") AND NOT ("intelligent tutoring system").

We included peer-reviewed papers published on topics related to literacy and education on AI at school. Then we excluded papers whose usage of AI was limited to 1) supporting computer-based learning only, with no focus on learning about AI; 2) supporting assessment/tutoring based on AI. We also excluded papers that targeted college students and those that were limited to K-12 programming/CS concepts as a prerequisite for learning about AI in the future. Following these inclusion and exclusion criteria, our search in Scopus returned an initial list of 750 documents. After we inspected the title, abstract, keywords and full-text screening, we obtained a final list of 179 documents.

Data collection extraction and synthesis strategy

Data collection extraction was performed, discussed, and coordinated through regular meetings. After inspecting and discussing 10% of the papers over multiple meetings, the authors agreed on the annotations presented in Table 2. This process is important as it allowed us to build a data annotation scheme empirically emerging from the sampled papers. A copy of the papers was also kept for easy review in case of doubts or disagreements.

The data resulted in a spreadsheet with the metadata of the papers which passed the inclusion and exclusion criteria, and a document with the list of paper IDs together with the rest of annotations. Some Python scripts were used to further process metadata (e.g., counting participating countries, frequencies, etc.) and produce a more complete bibliographic report with histograms and overview counting. A more qualitative analysis was carried out to answer the research question based on paper reading and annotations.

Results

The results were organized into two subsections. The first subsection is a bibliometric analysis of the reviewed studies, which is based on the metadata provided by Scopus. The second subsection provides a qualitative analysis of the studies, which is based on the

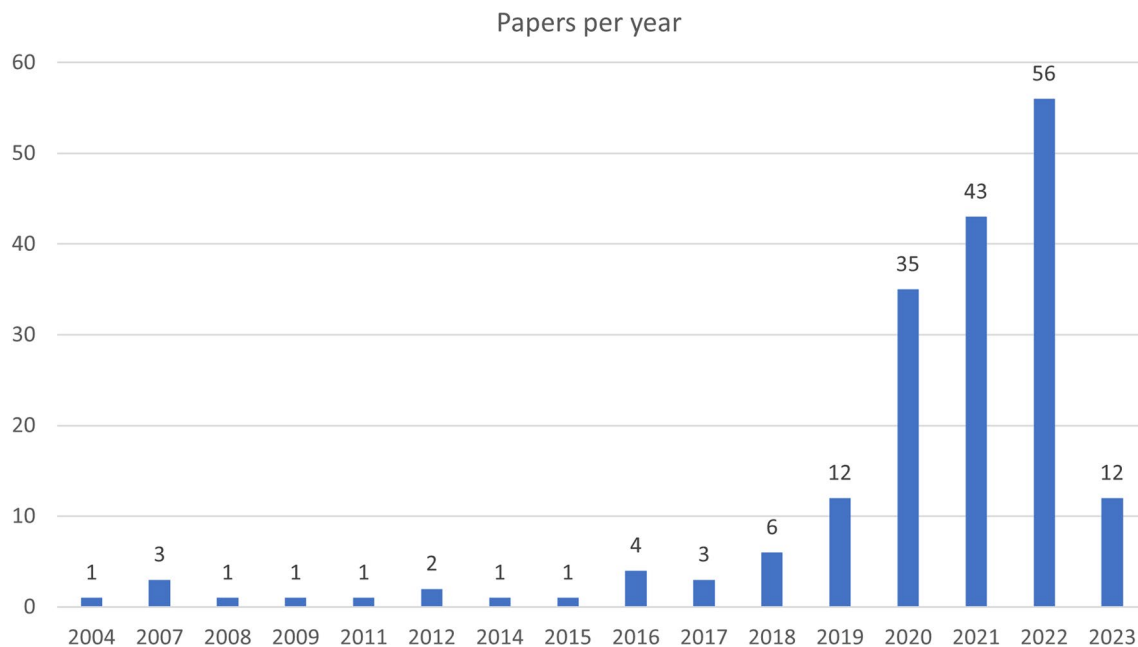


Fig. 1 Annual scientific production: number of papers by year

extracted data annotations (see Table 2). Both analyses are complementary and together deliver a better understanding of the research articles retrieved.

Bibliometric analysis

Figure 1 shows that the annual scientific production has been modest. It gained traction in 2016 and increased sharply in 2020.

Most of the contributions are conference publications (126 papers), while 52 are journal articles and one is a book chapter (Fig. 2).

Eighty out of 179 papers have at least a citation in Scopus. There are 13 papers that have 10 or more citations, and the most cited papers are Long and Magerko (2020) and Touretzky et al. (2019b). Figure 3 summarizes the number of contributions by publishers, where Springer, IEEE and ACM stand out, followed by Elsevier. As for journals, there are no single journals concentrating the publication of articles. Nevertheless, there are some journals that are especially relevant and well-known by the community such as the *International Journal of Child-Computer Interaction*, *Computers and Education: Artificial Intelligence*, *International Journal of Artificial Intelligence in Education*, or *IEEE Transactions on Education*.

As for conferences, Fig. 4 summarizes the main conference events where papers are published. It includes flagship conferences¹ such as CHI and AAAI, top-ranked

conferences such as HRI or SIGCSE and several noteworthy events (IDC, ICALT, ITiCSE, VL/HCC, to name a few). It is worth mentioning that AAAI is receiving contributions from recent years, which confirms the interest in the field in broadening the discussion to education. There are some additional publications associated with satellite AAAI events, such as workshops in CEUR-WS that deal with the issue under study. Although such contributions may sometimes be short, we decided to include them as they were relevant. For instance, the works published in (Herrero et al., 2020) and (Micheuz, 2020) include the German countrywide proposal for educating about AI, through a 6-module course focusing on explaining how AI works, the social discourse on AI and reducing existing misconceptions. On the other hand, Aguar et al. (2016) talk about teaching AI via an optional course which does not contribute to the final grades.

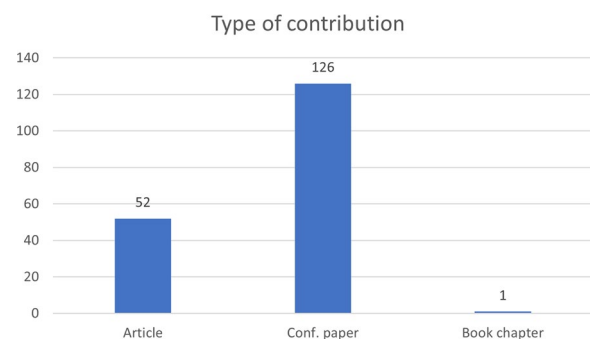


Fig. 2 Type of contributions: number of papers by type

¹ ¹Conference categorization and ranking based on the GII-GRIN-SCIE (GGS) Conference Ratings: <https://scie.lcc.uma.es/>

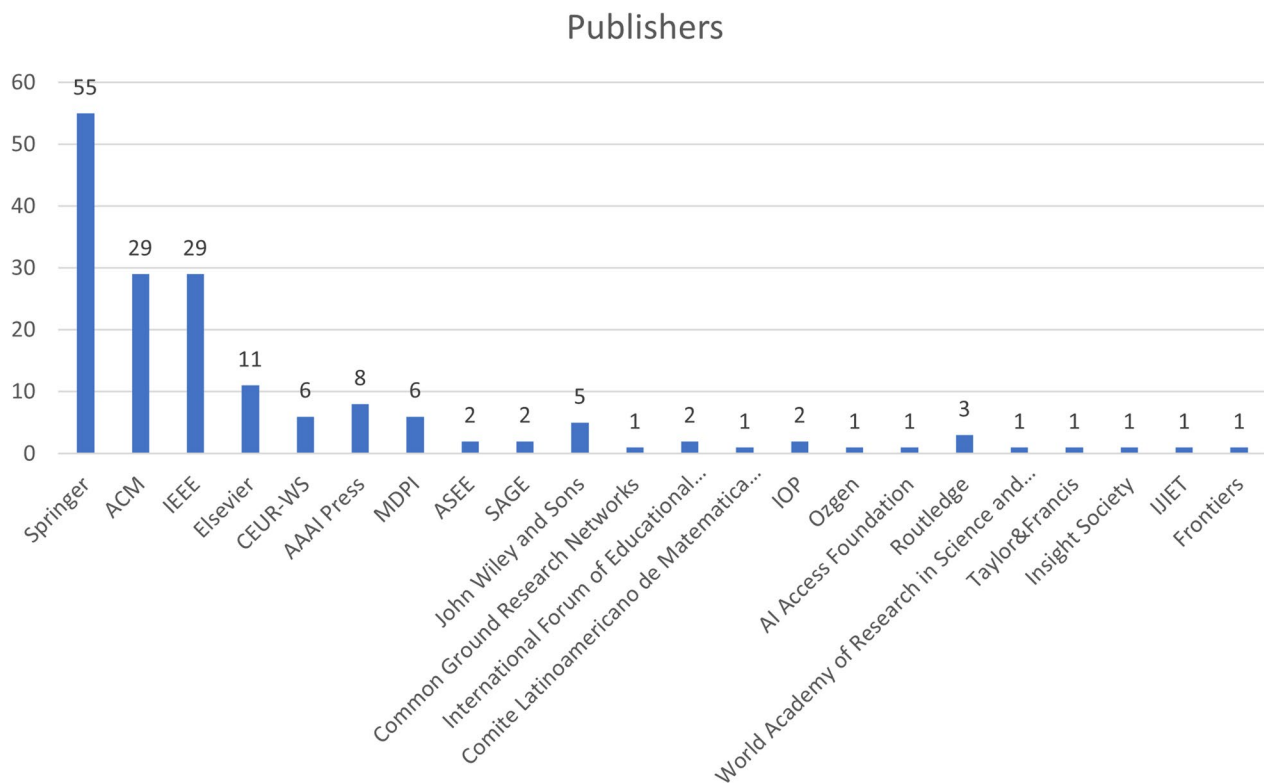


Fig. 3 Frequency of publishers: number of papers by publisher

The analysis did not reveal particularly outstanding institutions (see Table 3 for a summary). Among the 299 affiliated institutions, we mostly find universities and research centers along with a few collaboration associations. The most active institutions are the Chinese University of Hong Kong, University of Eastern Finland and MIT, whose authors participated in a total of 19, 11 and 10 contributions, respectively.

Finally, the retrieved papers were co-authored by 643 different authors affiliated to research institutions from 42 countries. Figure 5 shows the histogram of participation by country. Of the 179 papers reviewed, most papers were written by authors affiliated with institutions in the same country. Only 32 papers involved authors from several countries. It is remarkable that in these cases at least one author is from the US, Hong Kong or China.

Literature analysis

By analyzing the data extracted, the papers were classified into two broad thematic categories according to the type of educational approach, namely, learning experience and theoretical perspective. The first category covers AI learning experiences focused on understanding a particular AI concept/technique or using specific tools/platforms to illustrate some AI concepts. The second

category involves initiatives for the implementation of AI education for K-12 through the development of guidelines, curriculum design or teacher training, among others. Each main category was further subdivided into other subcategories to structure the field and characterize the different approaches used in developing AI literacy in K-12. Figure 6 shows all the identified categories and subcategories.

Learning experiences focused on understanding AI

This category covers learning experiences aimed at experimenting and becoming familiar with AI concepts and techniques. Based on the priority axes in AI literacy (Long & Magerko, 2020; Miao et al., 2021), we identified experiences aimed at acquiring basic AI knowledge to recognize artifacts using AI, learning how AI works, learning tools for AI and learning to live with AI.

Learning to recognize artifacts using AI

This subcategory refers to experiences that aim to understand AI concepts and techniques enabling the recognition of which artifacts/platforms use AI and which do not. Four studies were found in this subcategory. They are proposals aimed at helping young people to understand and demystify AI through different types of

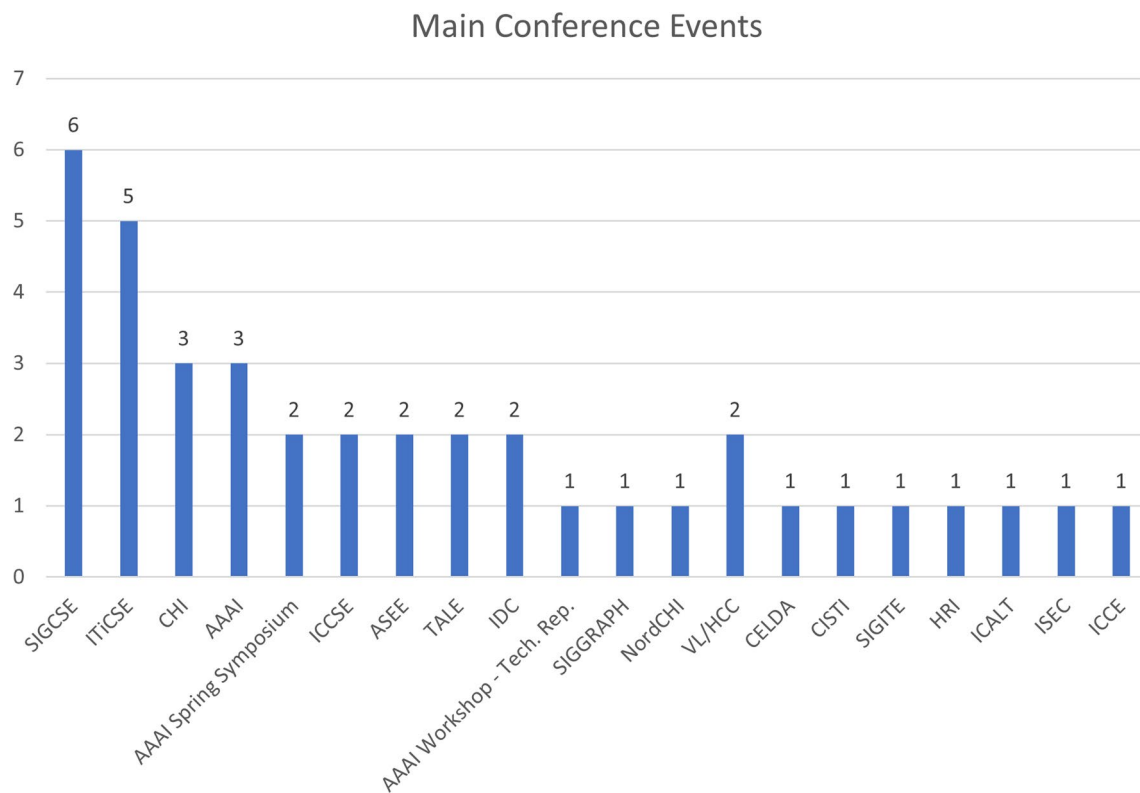


Fig. 4 Main conference events: number of papers by conference

Table 3 Most active institutions

Institutions	Papers
The Chinese University of Hong Kong (Hong Kong)	19
University of Eastern Finland (Finland)	11
MIT (USA)	10
North Carolina State University (USA)	8
Beijing Normal University (China)	6
Carnegie Mellon University (USA)	6
Indiana University (USA)	5
University of California (USA)	5
University of Florida (USA)	5
University of Southern California (USA)	5
Graz University of Technology (Austria)	4
South China Normal University (China)	4
Austrian Computer Society (Austria)	3
Georgia Institute of Technology (USA)	3
Korea University (South Korea)	3
18 institutions with 2 papers, and the rest with 1	–

activities. These activities included conducting discussions after watching AI-related movies (Tims et al., 2012), carrying out computer-based simulations of human-like behaviors (Ho et al., 2019), experimenting as active users

of social robots (Gonzalez et al., 2017) and programming AI-based conversational agents (Van Brummelen et al., 2021b).

Learning about how AI works

This topic covers proposals designed to understand how AI works to make user interaction with AI easier and more effective. In this type of proposal, the focus is on methodology and learning is achieved through technology (Kim et al., 2023). The objective is to provide a better understanding of a particular aspect of reality in order to carry out a project or solve a problem (Lenoir & Hasni, 2016). The activities are supported by active experiences based on building and creating intelligent devices to achieve the understanding of AI concepts following the idea of Papert's constructionism.

These experiences are mainly focused on teaching AI subfields such as ML or AI algorithms applied to robotics. Understanding the principles of ML, its workflows and its role in everyday practices to solve real-life problems has been the main objective of some studies (Burgsteiner et al., 2016; Evangelista et al., 2019; Lee et al., 2020; Sakulkeakulsuk et al., 2019; Vartiainen et al., 2021). In addition, there are also experiences focused on unplugged activities that simulate AI algorithms. For

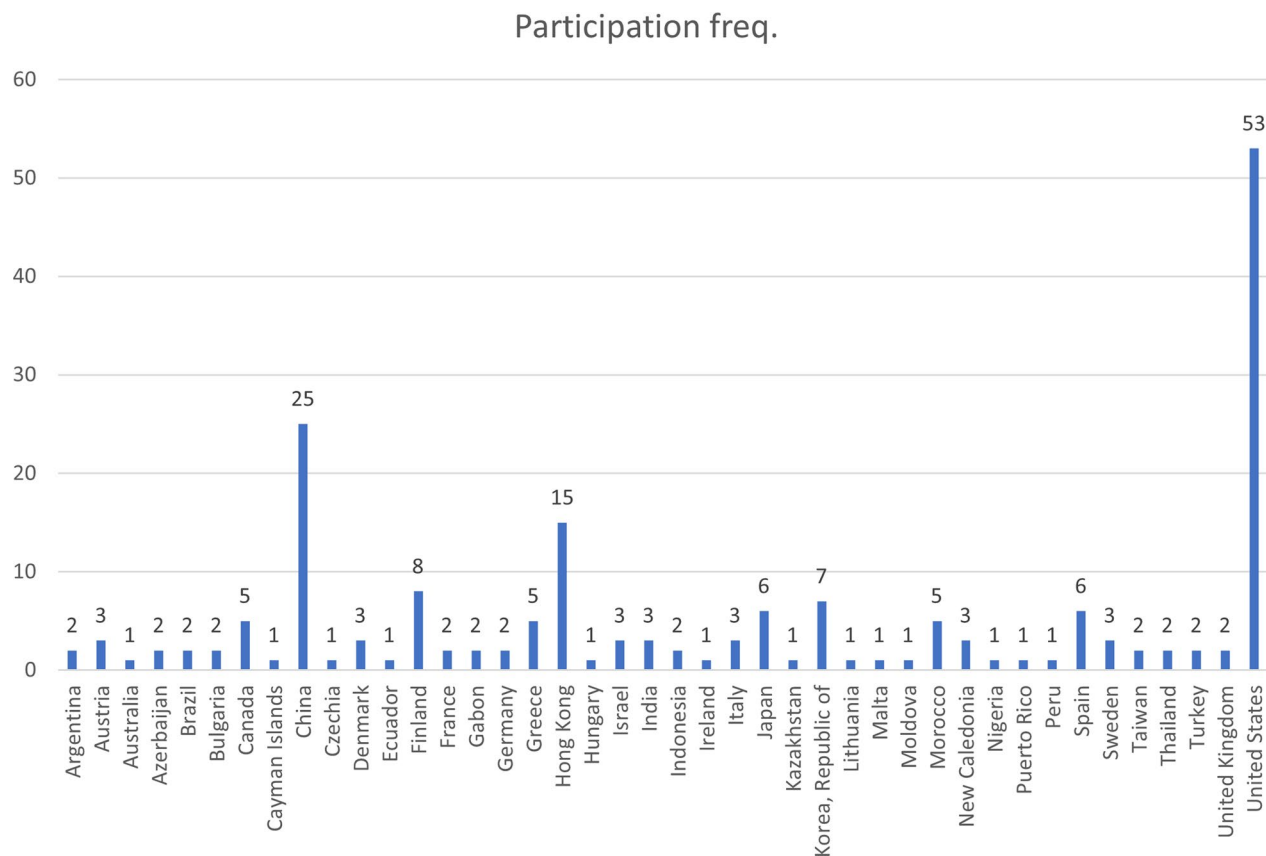


Fig. 5 Country participation: number of papers by country

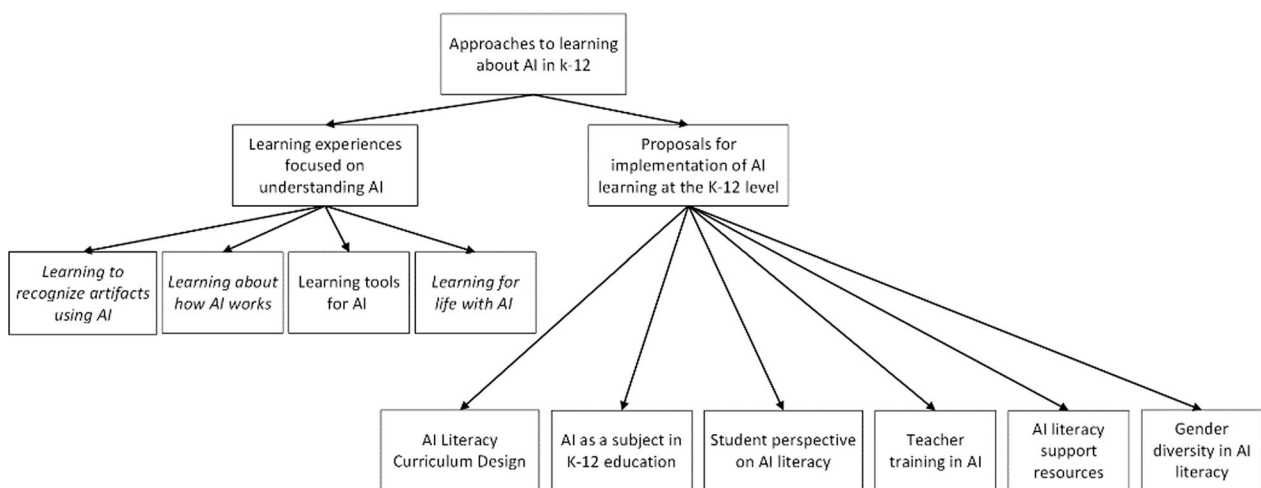


Fig. 6 Taxonomy of approaches to AI learning in K-12

example, through classic games such as Mystery Hunt, one can learn how to traverse a graph without being able to see beyond the next path to be traversed (blind search)

(Kandlhofer et al., 2016). Similarly, the AI4K12 initiative (Touretzky et al., 2019b) collects a large set of activities and resources to simulate AI algorithms.

Learning tools for AI

This topic includes approaches that involve learning about AI support tools. The development of intelligent devices in the context of teaching AI requires specific programming languages or age-appropriate tools. Many of the tools currently available are focused on ML, with the aim of demystifying this learning in K-12 education (Wan et al., 2020). Some of them are integrated into block-based programming languages (such as Scratch or App Inventor) (Toivonen et al., 2020; von Wangenheim et al., 2021), enabling the deployment of the ML models built into games or mobile applications. Other approaches use data visualization and concepts of gamification to engage the student in the learning process (Reyes et al., 2020; Wan et al., 2020) or combine traditional programming activities with ML model building (Rodríguez-García et al., 2020).

This type of proposal aims to introduce AI through tools that enable the use of AI techniques. It is therefore an approach focused on learning by using AI-oriented tools. In this vein, different experiences have focused on learning programming tools for applications based on Machine Learning (Reyes et al., 2020; Toivonen et al., 2020; von Wangenheim et al., 2021; Wan et al., 2020), robotics (Chen et al., 2017; Eguchi, 2021; Eguchi & Okada, 2020; Holowka, 2020; Narahara & Kobayashi, 2018; Nurbekova et al., 2018; Verner et al., 2021), programming and the creation of applications (Chittora & Baynes, 2020; Giannakos et al., 2020; Kahn et al., 2018; Kelly et al., 2008; Park et al., 2021). Some of these tools use Scratch-based coding platforms to make AI-based programming attractive to children. In (Kahn et al., 2018), students play around with machine learning to classify self-captured images, using a block-based coding platform.

There are also experiences in which other types of environments are used to facilitate learning (Aung et al., 2022). In (Holowka, 2020; Verner et al., 2021), students can learn reinforcement learning through online simulation. In (Narahara & Kobayashi, 2018), a virtual environment helps students generate data in a playful setting, which is then used to train a neural network for the autonomous driving of a toy car-lab. In (Avanzato, 2009; Croxell et al., 2007), students experiment with different AI-based tasks through robotics-oriented competitions.

Learning for life with AI

This subcategory covers experiences aimed at understanding how AI can affect our lives thus providing us with skills to critically assess its technology. In (Vachovsky et al., 2016), technically rigorous AI concepts are contextualized through the impact on society. There are also experiences where students explore how a robot

equipped with AI components can be used in society (Eguchi & Okada, 2018), program conversational agents (Van Brummelen et al., 2021b), or learn to recognize credible but fake media products (video, photos), which have been generated using AI-based techniques (2021b; Ali et al., 2021a).

The ethical and philosophical implications of AI have also been addressed in some experiences (2021b; Ali et al., 2021a; Ellis et al., 2005), whereas others focus on training students to participate in present-day society and become critical consumers of AI (Alexandre et al., 2021; Cummings et al., 2021; Díaz et al., 2015; Kaspersen et al., 2022; Lee et al., 2021; Vartiainen et al., 2020).

Proposals for implementation of AI learning at the K-12 level

Some countries are making efforts to promote AI education in K-12. In the U.S., intense work is being carried out on the integration of AI in schools and among these schemes, AI4K12 stands out (Heintz, 2021). This scheme is especially interesting since it defines the national guidelines for future curricula, highlighting the essential collaborative work between developers, teachers and students (Touretzky et al., 2019a). This idea of co-creation is also stressed in other schemes (Chiu, 2021). In the U.S. we can also mention the proposal made by the Massachusetts Institute of Technology, which is an AI curriculum that aims to engage students with its social and ethical implications (Touretzky et al., 2019a). Although the United States is working intensively on the design of integrating this knowledge into the curriculum, so far AI is not widely offered in most K-12 schools (Heintz, 2021).

In China, the Ministry of Education has integrated AI into the compulsory secondary school curriculum (Ottenbreit-Leftwich et al., 2021; Xiao & Song, 2021). Among their schemes we can reference the AI4Future initiative of the Chinese University of Hong Kong (CUHK), which promotes the co-creation process to implement AI education (Chiu et al., 2021). In Singapore, a program for AI learning in schools has also been developed, where K-12 children learn AI interactively. However, the program is hindered by a lack of professionals (teachers) with adequate training (Heintz, 2021). In Germany, there are also several initiatives to pilot AI-related projects and studies (Micheuz, 2020), including the launch of a national initiative to teach a holistic view of AI. This initiative consists of a 6-module course aimed at explaining how AI works, stimulating a social discourse on AI and clarifying the abundant existing misconceptions (Micheuz, 2020). Canada has also designed an AI course for high schools. The course is intended to empower students with knowledge about AI, covering both its philosophical and conceptual underpinnings

as well as its practical aspects. The latter are achieved by building AI projects that solve real-life problems (Nisheva-Pavlova, 2021).

The literature also highlights the different approaches that AI literacy should focus on: curriculum design, AI subject design, student perspective, teacher training, resource design and gender diversity. All these approaches are described in depth below.

AI literacy curriculum design

Approaches to curriculum development differ widely, ranging from the product-centered model (technical-scientific perspective) to the process-centered model (learner perspective) (Yue et al., 2021). AI literacy can be launched in primary and secondary education depending on the age and computer literacy of the students. To do this, it is necessary to define the core competencies for AI literacy according to three dimensions: AI concepts, AI applications and AI ethics and security (Long & Magerko, 2020; Wong et al., 2020). Research has focused on the understanding of the concepts, the functional roles of AI, and the development of problem-solving skills (Woo et al., 2020). This has led to proposing a redefinition of the curriculum (Han et al., 2019; Malach & Vicherková, 2020; Zhang et al., 2020) supported by different ideas that K-12 students should know (Chiu et al., 2021; Sabuncuoglu, 2020; Touretzky et al., 2019b). Several countries have already made different curricular proposals (Alexandre et al., 2021; Micheuz, 2020; Nisheva-Pavlova, 2021; Ottenbreit-Leftwich et al., 2021; Touretzky et al., 2019b; Xiao & Song, 2021), where they argue that the curricular design must include different elements such as content, product, process and praxis (Chiu, 2021). It is also convenient for learning in AI to follow the computational thinking model (Shin, 2021), contextualizing the proposed curriculum (Eguchi et al., 2021; Wang et al., 2020) and providing it with the necessary resources for teachers (Eguchi et al., 2021). In this sense, emerging initiatives highlight the need to involve teachers in the process of co-creating a curriculum associated to their context (Barlex et al., 2020; Chiu et al., 2021; Dai et al., 2023; Lin & Brummelen, 2021; Yau et al., 2022).

AI as a subject in K-12 education

Traditionally, including computer science or new technologies in the educational system has been carried out through a specific subject integrated into the curriculum or through the offer of extracurricular activities. In this sense, different proposals have suggested the integration of AI as a subject in K-12 education (Ellis et al., 2009; Knijnenburg et al., 2021; Micheuz, 2020; Sperling & Lickerman, 2012), in short-term courses (around 15 h) and divided into learning modules focused on classical and

modern AI (Wong, 2020) or through MOOCs (Alexandre et al., 2021).

Student perspective on AI Literacy

Student-focused studies explore and analyze attitudes and previous knowledge to make didactic proposals adapted to the learner. Some of them measure their intention and interest in learning AI (Bollin et al., 2020; Chai et al., 2021, 2020a, 2020b; Gao & Wang, 2019; Harris et al., 2004; Sing, et al., 2022; Suh & Ahn, 2022), whereas others discuss their views on the integration of technologies in the education system (Sorensen & Koefoed, 2018) and on teaching–learning support tools in AI (Holstein et al., 2019).

Teacher training in AI

Teachers are key players for the integration of AI literacy in K-12, as proven by the numerous studies that examine this issue (An et al., 2022; Bai & Yang, 2019; Chiu & Chai, 2020; Chiu et al., 2021; Chounta et al., 2021; Judd, 2020; Kandlhofer et al., 2019, 2021; Kim et al., 2021; Korenova, 2016; Lin et al., 2022; Lindner & Berges, 2020; Oh, 2020; Summers et al., 1995; Wei et al., 2020; Wu et al., 2020; Xia & Zheng, 2020). This approach places teachers at the center, bearing in mind what they need to know so as to integrate AI into K-12 (Itmazi & Khlaif, 2022; Kim et al., 2021). The literature analyzed reports on the factors that influence the knowledge of novice teachers (Wei, 2021) and focuses on teacher training in AI (Lindner & Berges, 2020; Olari & Romeike, 2021). Thus, AI training proposals can be found aimed at both teachers in training (Xia & Zheng, 2020) and practicing educators. Training schemes focus on their knowledge in technologies to facilitate their professional development (Wei et al., 2020) through the TPACK (Technological, Pedagogical and Content Knowledge) teaching knowledge model (Gutiérrez-Fallas & Henriques, 2020). Studies focusing on teachers' opinions on curriculum development in AI are relevant (Chiu & Chai, 2020), as are their self-efficacy in relation to ICT (Wu et al., 2020), their opinions on the tools that support the teaching–learning process in AI (Holstein et al., 2019) and their teacher training in technologies (Cheung et al., 2018; Jaskie et al., 2021). These elements are central to the design of an AI literacy strategy in K-12. Both the co-design of ML curricula between AI researchers and K-12 teachers, and the assessment of the impact of these educational interventions on K-12 are important issues today. At present, there is a shortage of teachers with training in AI and working with teachers in training (Xia & Zheng, 2020) or with teachers in schools (Chiu et al., 2021) is proposed as an effective solution. One of the most interesting analyses of teacher competency

proposes the acquisition of this skill for the teaching of AI in K-12, through the analysis of the curricula and resources of AI using TPACK. This model was formulated by (Mishra & Koehler, 2006) and aims to define the different types of knowledge that teachers need to integrate ICT effectively in the classroom. In this regard, it is suggested that teachers imparting AI to K-12 students require TPACK to build an environment and facilitate project-based classes that solve problems using AI technologies (Kim et al., 2021).

AI literacy support resources

Research using this approach focuses on presenting resources that support AI literacy (Kandlhofer & Steinbauer, 2021), considering that the creation of resources and repositories is a priority in supporting this teaching-learning process (Matarić et al., 2007; Mongan & Regli, 2008). However, these resources largely do not meet an interdisciplinary approach and do not embody a general approach to AI development (Sabuncuoglu, 2020).

Gender diversity in AI literacy

AI education, as a broad branch of computer science, also needs to address the issue of gender diversity. Lack of gender diversity can impact the lives of the people for whom AI-based systems are developed. The literature highlights the existence of proposals designed with a perspective toward gender, where the activities designed are specifically aimed at girls (Ellis et al., 2009; Jagannathan & Komives, 2019; Perlin et al., 2005; Summers et al., 1995; Vachovsky et al., 2016; Xia et al., 2022).

Discussion

The huge impact that AI is having on our lives, at work and in every type of organization and business sector is easily recognizable today. No one doubts that AI is one of the most disruptive technologies in history, if not the most. In recent years, the expectations generated by AI, far from being deflated, have only grown. We are still a long way from general-purpose AI, but the application of AI to solve real problems has already taken hold for a wide range of purposes. It is therefore necessary for young people to know how AI works, as this learning will make it easier for them to use these technologies in their daily lives, both to learn and to interact with others.

Like any other technology, the potential uses and abuses of AI go hand in hand with its disruptive capacity. Many social groups and governments are expressing concern about the possible negative consequences of AI misuse. Although it is crucial to adequately regulate the use of AI, education is as important, if not more important, than regulation. Everything, whether good or bad, stems

from the education received. Thus, education systems must prepare students for a society in which they will have to live and interact with AI. AI education will enable young people to discover how these tools work and, consequently, to act responsibly and critically. Therefore, AI literacy has become a relevant and strategic issue (Chiu & Chai, 2020).

This systematic review has focused on analyzing AI teaching-learning proposals in K-12 globally. The results confirm that the teaching of basic AI-related concepts and techniques at the K-12 level is scarce (Kandlhofer et al., 2016). Our work shows that there have been, on the one hand, different AI learning experiences and, on the other hand, proposals for the implementation of AI literacy, made at the political level and by different experts. The learning experiences described show that AI literacy in schools has focused on technical, conceptual, and applied skills in some domains of interest. Proposals for AI implementation, especially those defined by the US and China, reveal that significant efforts are being made to design models that frame AI literacy proposals.

We also found that there are hardly any AI learning experiences that have analyzed learning outcomes, e.g., through assessments of learners' understanding of AI concepts. Obviously, this is a result of the infancy of these AI learning experiences at the K-12 level. However, it is important for learning experiences to be based on clearly defined competencies in a particular AI literacy framework, such as those proposed in the literature (Alexandre et al., 2021; Han et al., 2019; Long & Magerko, 2020; Malach & Vicherková, 2020; Micheuz, 2020; Ottenbreit-Leftwich et al., 2021; Touretzky et al., 2019a; Wong et al., 2020; Xiao & Song, 2021; Zhang et al., 2020). Recently, Van Brummelen et al. (2021a) designed a curriculum for a five-day online workshop based on the specific AI competencies proposed by Long and Magerko (2020). They used several types of questionnaires to assess the quality of the program through the knowledge acquired by the students in these competencies. Therefore, clearly defined competency-based learning experiences can provide a rigorous assessment of student learning outcomes.

The research shows that clear guidelines are needed on what students are expected to learn about AI in K-12 (Chiu, 2021; Chiu & Chai, 2020; Lee et al., 2020). These studies highlight the need for a competency framework to guide the design of didactic proposals for AI literacy in K-12 in educational institutions. This framework would provide a benchmark for describing the areas of competency that K-12 learners should develop and which specific educational projects can be designed. Furthermore, it would support the definition of a curriculum reflecting sequence and academic continuity (Woo et al., 2020). Such a curriculum should be modular and personalized

(Gong et al., 2019) and adjusted to the conditions of the schools (Wang et al., 2020). In the teaching of AI, an exploratory education should be adopted, which integrates science, computer science and integral practice (Wang et al., 2020). It should also address issues related to the ethical dimension, which is fundamental to the literacy of K-12 students as it enables them to understand the basic principles of AI (Henry et al., 2021). This training facilitates the development of students' critical capacity, and this is necessary to understand that technology is not neutral and to benefit from and make appropriate use of it. Ethics, complementary to legal norms, enhances the democratic quality of society by setting legitimate limits in the shaping of technological life. In this sense, different AI literacy proposals in K-12 already support the addressing of ethical, social and security issues linked to AI technologies (Eguchi et al., 2021; Micheuz, 2020; Wong et al., 2020). Moreover, considering designing for social good could foster or help to motivate learning about AI (Chai et al., 2021). Without a doubt, all this will impact on the achievement of a more democratic society. Due to the gender gap in issues related to computer science, it is also necessary to address the gender perspective. In this vein, the research proposes, among other strategies, to focus AI literacy on real-world elements since this approach favors the motivation of girls and greater involvement in learning (Jagannathan & Komives, 2019). However, little attention is paid to the undesirable consequences of an indiscriminate and insufficiently thought-out application of AI, both in higher education and especially in K-12. For example, the increase in socio-economic inequality between countries and within countries, resulting from the increasing automation of employment, is of particular concern. This is leading to growing inequality in wages and preservation of human employment, but it is not usually a subject of interest in education.

Currently, the challenges of this AI literacy require an interdisciplinary and critical approach (Henry et al., 2021). We believe that AI literacy can be leveraged to enhance the learning of disciplinary core subjects by integrating AI into the teaching process of those subjects. AI literacy should rely on transferring AI knowledge and methods to core subjects, allowing education to cross disciplinary boundaries, but staying within the framework of disciplinary core subjects. To achieve this change, educators need to take a closer look at the current capabilities of AI. This would enable them to identify all options to improve the core of educational practice and thus optimize the educational process. For example, understanding and using word clouds is a powerful educational strategy to enhance education in core subjects such as science (e.g., to facilitate object

classification), language (e.g. to enable the matching of different topics or authors' works), music (e.g., to support the analysis of song lyrics) or social sciences (e.g., to assist in comparing different discourses). Since AI is highly interdisciplinary in nature, it has a broad projection on multiple fields and problems that require a transversal and applied approach. For example, the basic algorithms of ML could be taught in Mathematics and related disciplines, the design of supervised classifiers could be performed for the study of taxonomies in Biology, natural language processing could be used to make the study of a language more attractive, or the ethical issues surrounding AI could be discussed in Philosophy and Social Sciences subjects.

Finally, for this meaningful learning to take place, AI teaching must be addressed through holistic, active, and collaborative pedagogical strategies in which real problem solving is the starting point of the learning process. An important gap regarding the integration of AI in K-12 concerns teachers, as it is unclear how to prepare and involve them in the process (Chiu & Chai, 2020). Teachers' attitudes towards AI have a significant influence on the effectiveness of using AI in education. Teachers can swing between total resistance and overconfidence. The first could arise from inadequate, inappropriate, irrelevant, or outdated professional development. On the one hand, teachers must be digitally-competent enough to integrate AI into the teaching-learning processes of their subjects. Therefore, teacher training is also necessary following a framework of standard competencies. This should include new ways of organizing the professional role of teachers, as well as enhancing students' attitudes towards these changes. On the other hand, research reveals that it is essential for didactic proposals to be co-designed and implemented by the teachers at those schools involved (Henry et al., 2021), to undergo training in the specific AI subjects and for this knowledge to be integrated into non-computer subjects (Lin & Brumelen, 2021). To this end, it is crucial to identify the perception and knowledge that teachers have about AI and involve them in the design of curricular proposals (Chiu, 2021; Chiu & Chai, 2020; Chiu et al., 2021).

Conclusion

This study aimed to understand how AI literacy is being integrated into K-12 education. To achieve this, we conducted a search process following the systematic literature review method and using Scopus. Two broad groups of AI literacy approaches were identified, namely learning experiences and theoretical perspective. The study revealed that learning experiences in schools have focused mainly on technical and applied skills limited to a specific domain without rigorously assessing student

learning outcomes. In contrast, the US and China are leading the way in AI literacy implementation schemes which are broader in scope and involve a more ambitious approach. However, there is still a need to test these initiatives through comprehensive learning experiences that incorporate an analysis of learning outcomes. This work has allowed us to draw several conclusions that can be considered in the design of AI literacy proposals in K-12. Firstly, AI literacy should be based on an interdisciplinary and competency-based approach and integrated into the school curriculum. There is no need to include a new AI subject in the curriculum, but rather to build on the competencies and content of disciplinary subjects and then integrate AI literacy into those subjects. Given the interdisciplinary nature of AI, AI education can break disciplinary boundaries and adopt a global, practical, and active approach in which project-based and contextualized work plays an important role. Secondly, AI literacy should be leveraged to extend and enhance learning in curricular subjects. As a final point, AI literacy must prioritize the competency of teachers and their active participation in the co-design of didactic proposals, together with pedagogues and AI experts.

Supplementary Information

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Additional file 1. Additional listing.

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Author contributions

All authors have contributed significantly to the authorship of this work in all stages of conceptualization, discussions, definition of the methodology, carrying out the analysis as well as writing—review & editing. All authors read and approved the final manuscript.

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Availability of data and materials

Last revision round required update the review. Thus, Additional file 1 contains a.csv file with the listing of papers that are not cited but are part of the reviewed papers. The papers cited in text already appear in the Reference section and, therefore, not in the Additional file.

Declarations

Ethics approval and consent to participate

This research is carried out in accordance to ethics recommendations. As it focuses on a systematic literature review as a research method, ethics approval by the University ethics committee does not apply.

Competing interests

The authors declare that they have no competing interests. The authors have no other relevant financial or non-financial interests to disclose and no further competing interests to declare that are relevant to the content of this article.

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